

Week 2 - Assignment 4: Prompt Ladder

Objective: Improve one weak prompt through five iterations, changing exactly one element at a time and observing how the output changes.

Baseline Prompt

Prompt:

Help me frame my machine learning task.

Representative Output:

Identify the ML task, define a target, and choose a metric.

What changed: Initial weak prompt.

What improved in the output: No improvement (baseline).

What still failed: Too generic and lacks context.

What I would try next: Add a clearer goal.

Version 1 - Clear Goal

Prompt:

Help me frame my Content Refresh Prioritization project as a machine learning task.

Representative Output:

Identified the project as a ranking problem.

What changed: Added a clear goal.

What improved in the output: The response became project-specific.

What still failed: Still lacked enough context.

What I would try next: Add project context.

Version 2 - Context

Prompt:

Help me frame my Content Refresh Prioritization project. It uses clicks, impressions, CTR, and average position to prioritize webpages.

Representative Output:

Suggested ranking pages using multiple search signals.

What changed: Added project context.

What improved in the output: Recommendations became more specific.

What still failed: Output was not well organized.

What I would try next: Specify an output format.

Version 3 - Output Format

Prompt:

Present the answer as a table with Business Problem, ML Task, Target, Success Metric, and Why ML.

Representative Output:

Returned a structured table.

What changed: Specified output format.

What improved in the output: The answer became easier to read.

What still failed: Some explanations remained generic.

What I would try next: Define the audience.

Version 4 - Audience

Prompt:

Write the explanation for FlyRank ML Internship mentors reviewing student work.

Representative Output:

Produced a more professional explanation.

What changed: Added target audience.

What improved in the output: Writing became more suitable for submission.

What still failed: Still contained some generic wording.

What I would try next: Add quality criteria.

Version 5 - Quality Criteria

Prompt:

Keep the explanation concise, practical, avoid buzzwords, justify recommendations, and explain why ranking beats rules.

Representative Output:

Produced a concise and well-justified explanation.

What changed: Added quality criteria.

What improved in the output: The answer became directly reusable.

What still failed: Could include more project-specific examples.

What I would try next: Use this as the final prompt.

Final Reusable Prompt

Help me frame my Content Refresh Prioritization project for a FlyRank ML Internship submission.

Project Context: The project analyzes search performance metrics such as clicks, impressions, CTR, and average position to prioritize webpages for content refresh.

Requirements: Identify the ML task type. Define a prediction target or proxy. Recommend an evaluation metric. Explain why ML is more suitable than a fixed rule. Present the answer in a clear table. Keep the explanation concise and practical. Avoid generic AI language.